



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
ISE-I (Semester-I) 2025-26

Class: First Year M. Tech.

Branch: All branches

Course: Open Elective: Operations Research

Course Code: OEC-803

Date: 17-Oct-2025

Time: 16:30 to 17:15

Max. Marks: 20

Q. No.	Questions	Marks	CO	BT
Q. 1)	Define systems engineering. Differentiate between systems approach and system analysis. How is it related to operations research?	07	1	1
Q. 2)	What is a model? Explain in brief role of systems models and state the different types of system models.	07	3	2
Q. 3)	Explain the concept of system with a neat diagram.	06	4	2

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	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Mechanical Engineering In-Semester Examination I Academic Year: 2025-26 (Semester – I)		
Class: FY M.Tech.	Subject: Computer Aided Design	Subject Code: PEC-PL504	
Date: 14-10-2025	Time: 16:30 to 17:15	Max. Marks: 20	

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Question	Marks	CO	BT
1	Discuss the use of computer in assisting design process	8	CO1	BT3
2	Discuss form and function	4	CO2	BT1
3	Differentiate CPU and GPU	4	CO1	BT3
4	Write a short note on graphics pipeline	4	CO1	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Classify CAD hardware and software for variety of applications.
CO 2	Exhibits importance and use of various CAD modelling techniques.
CO 3	Exhibits the idea and procedure of data transfer in various CAD tools.
CO 4	Calculate the position, shape and geometry of CAD model after translation or transformation
CO5	Demonstrate the assembly procedure of CAD applications.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
In-Semester Examination I
Academic Year: 2025-26 (Semester -I)

Class: M.Tech.	Subject: Computational Tools and Programming - I	Subject Code: PEC-PL-505
Date: 15.10.2024	Time: 16:30 to 17:15	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. This paper contains one page.

Q.N.		Marks	CO	BT
Q.1	Explain the Pointer, Reference Operator, and Dereference operator using a simple program.	6	CO1	BT2
Q.2	Write the output of programs given below; Program1 1 #include <iostream> 2 using namespace std; 3 int x=20; 4 int main() 5 { 6 int x=10; 7 { 8 int x=30; 9 cout<<x; 10 } 11 return 0; 12 } Program2 1 #include <iostream> 2 using namespace std; 3 int x=20; 4 int main() 5 { 6 int x=10; 7 { 8 //int x=30; 9 cout<<x; 10 } 11 return 0; 12 }	2	CO1	BT3
Q.3	Write a program to add two 3x3 matrices using a for loop.	6	CO1	BT3
Q.4	Write a C++ program to calculate the factorial of a given number using recursive function.	6	CO1	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT): BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Write programs in languages like C, C++, C#, VB.
CO 2	Work in software development projects.
CO 3	Effectively use computing tools like spreadsheets, Octave for data processing, engineering calculations and write programs therein.
CO 4	Write algorithms and programs for development of engineering applications.

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Mechanical Engineering In-Semester Examination I Academic Year: 2025-26 (Semester – I)		
Class: First Year M.Tech.	Subject: Data Management (DM)	Subject Code: PCC-PL-503	
Date: 13/10/2025	Time: 16:30 to 17:15	Max. Marks: 20	

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/sharing of stationery, calculator, etc. is not allowed.
- Figures to the right indicate full marks.
- The symbols carry the usual meaning.

Q.N.			Marks	CO	BT
Q.1		Solve any TWO .			
	a	Explain the advantages of Database Management Systems over the File System.	05	CO1	BT2
	b	What is Database Transaction? Draw a Data Table that explain the Transaction Management.	05	CO1	BT1
	c	Explain the Role Database Administrators in an Organization.	05	CO1	BT2
Q.2		Solve any TWO .			
	a	Write an overview of the design process in DBMS along with the database design models.	05	CO2	BT2
	b	Draw and explain all the Symbols used in E-R diagram.	05	CO2	BT2
	c	Create the Entity Relationship Diagram with below requirements:- 1. Company is organized in DEPARTMENTS . Each department has a unique Name, Number and a particular employee who manages the department also keep the track of start date of the manager. 2. A department controls number of PROJECTS , each of which has a unique number and single Location. 3. EMPLOYEE Details - Name, SSN, Sex, Salary, also keep track of number of hours per week on each project. 4. Keep track of each employee's DEPENDENTS (FName, MName, LName, Sex, Relationship to the Employee)	05	CO2	BT6

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remembering, BT2 - Understanding, BT3 - Applying, BT4 - Analyzing, BT5 - Evaluating, BT6 - Creating

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1.	Demonstrate program Data independence, Data models for database system, Database Schema and Database Instances.
CO2.	Construct Structured Query Language Statements used in creation and manipulation of database.